Network communication technologies

L17
FACT FILE

Specification: students should be able to

- Describe Wi-Fi, Bluetooth, optical fibre and mobile communication technology (4G and 5G)
- Evaluate the effectiveness of each communication technology in terms of speed, range, reliability, security and cost
- Compare wireless communication with optical fibre communication
- Select and justify a suitable communication technology for a given scenario

CCEA fact file: Network Technologies 2

Key words: Wi-Fi, wireless router, radio waves, Mbps, Bluetooth, pairing, optical fibre, bandwidth, repeater, interference, 4G, 5G, latency, coverage, encryption



Typical range of Bluetooth, Wi-Fi, 4G, 5G and optical fibre

Overview of communication technologies

Networks can send data along cables or through the air. Wi-Fi uses radio waves and a wireless router so devices can join a network without cables. Bluetooth also uses radio signals, but it links two nearby devices directly, over about 10 metres, after they have been paired. Optical fibre sends data as pulses of light along thin glass strands, giving very high bandwidth over long distances. Mobile technologies, 4G and 5G, use radio masts to give wireless internet across a wide area, so people can stay connected while they move. To evaluate these technologies, think about five things: speed (bandwidth), range, reliability, security and cost. No single technology is best for everything, so the right choice depends on the situation.

Wi-Fi and Bluetooth

Wi-Fi lets laptops, phones and smart devices connect to a network through a wireless router or access point. It is cheap and quick to set up because no cables are needed, and users can move around a building. However, the signal gets weaker the further you are from the router, and walls and other devices cause interference. Speed also falls when many people share one access point. Older Wi-Fi standards reach about 54 Mbps, while modern standards are much faster. A Wi-Fi network should have a strong password and WPA2 or WPA3 encryption. Bluetooth is designed for short distances. It uses very little power and needs no router, which makes it ideal for earbuds, smartwatches, wireless mice and hands-free car kits. Its range is short, it is slower than Wi-Fi and only a few devices can connect at once.

Optical fibre

An optical fibre cable is a bundle of very thin glass strands. Data travels along each strand as pulses of light, and because each strand carries its own signal, a huge amount of data can be sent at the same time. This gives very high bandwidth and speed. Light is not affected by electrical interference, so fibre is reliable near machinery, and it is very hard for a hacker to tap into the cable without being noticed, so data is secure. Signals can travel many kilometres with little loss, although repeaters are needed to boost them over very long distances. The main drawback is cost: installing fibre often means digging up roads and paying specialists. The glass can also break if the cable is bent too sharply. Fibre is used for internet backbones, undersea cables and full-fibre broadband to homes.

4G and 5G mobile technology

Mobile phone networks divide an area into cells, each served by a mast. Phones send and receive data from the nearest mast using radio waves. 4G gives high-speed data, good enough for streaming HD video and making video calls, and it lowered the cost of mobile data. Because masts are expensive, coverage is best in towns and weaker in rural areas. Using mobile data also drains the battery faster than Wi-Fi. 5G is faster again and has much lower latency, which means less delay. It can support many more devices in one place, which helps the Internet of Things, cloud gaming and connected vehicles. The highest-speed 5G signals do not travel as far and are blocked by buildings, so many more masts are needed and coverage is still growing.

Key terms from the mark schemes

Wi-Fi
Wireless technology that uses radio waves and a wireless router or access point to connect devices to a network without cables.
Credited: Wireless fidelity; wireless LAN access

Bluetooth
Short-range, low-power wireless technology that connects two nearby devices directly after they have been paired.
Credited: Short-range wireless link between devices

Optical fibre
Cable made of thin glass strands that carries data as pulses of light, giving very high bandwidth and immunity to electrical interference.
Credited: Fibre optic

4G
Fourth-generation mobile technology giving high-speed wireless data through a network of masts.

5G
Fifth-generation mobile technology with faster speeds, much lower latency and capacity for more devices than 4G, but needing more masts.
Credited: Fast, low-latency mobile network

Bandwidth
The amount of data that can be transmitted in a given time, measured in bits per second.
Credited: Data transfer rate

Latency
The delay between data being sent and being received.
Credited: Lag

Pairing
Linking two Bluetooth devices so they recognise each other and can exchange data.
Credited: Synchronising

Repeater
A device that boosts a signal so it can travel over longer distances.

Interference
Unwanted signals or obstacles that weaken or corrupt transmitted data.

Exam focus

Questions on this lesson often ask you to match technologies to uses or descriptions, as in 2021 Q11, 2022 Q8(b) and 2023 Q6(a). Learn one clear feature and one typical use for each technology: Bluetooth for short-range links such as earbuds, Wi-Fi for wireless access to a network in a building, fibre for very fast secure links, 5G for fast, low-latency mobile data. In 2022 Q8(d) candidates judged true/false statements about fibre optic, so know the facts exactly: light pulses, glass strands, high bandwidth, not affected by electrical interference, hard to tap, expensive to install. When asked to evaluate or compare, give both an advantage and a disadvantage and use comparative words such as 'faster than' or 'more secure than'. Vague answers like 'it is better' or 'it is quick' earn nothing unless you say what it is compared with.

Unit 1 2023 Q6	[5]	Communication technologies; IoT
Unit 1 2022 Q8	[11]	IoT, Wi-Fi and Bluetooth, Internet vs WWW, fibre optic
Unit 1 2021 Q11	[8]	Key term matching

Check yourself

1. Describe Bluetooth technology and give one typical use. [3]
2. A family in Larne connect their phones, tablets and smart TV to the internet using Wi-Fi. State two features of the wireless technology they are using. [2]
3. Give two advantages and one disadvantage of optical fibre as a communications medium. [3]

Answers: 1. Short-range wireless technology (1) / uses radio signals to connect two devices directly (1) / devices must be paired/synchronised (1) / low power (1) / no router needed (1) – max 2 / use, e.g. wireless headphones, hands-free car kit, wireless mouse, smartwatch to phone (1) 2. Any two: uses radio waves (1) / needs a wireless router or access point (1) / no network cables required (1) / devices have a built-in wireless adapter (1) / signal strength depends on distance from the router (1) / should be protected by a password/encryption (1) / speed depends on the Wi-Fi standard (1) 3. Advantages (any two): very high bandwidth/speed (1) / not affected by electrical interference (1) / very hard to tap so data is secure (1) / carries data long distances with little loss (1) / Disadvantage (any one): expensive to install (1) / fragile – can break if bent (1) / needs repeaters over very long distances (1) / specialist skills needed to repair (1)